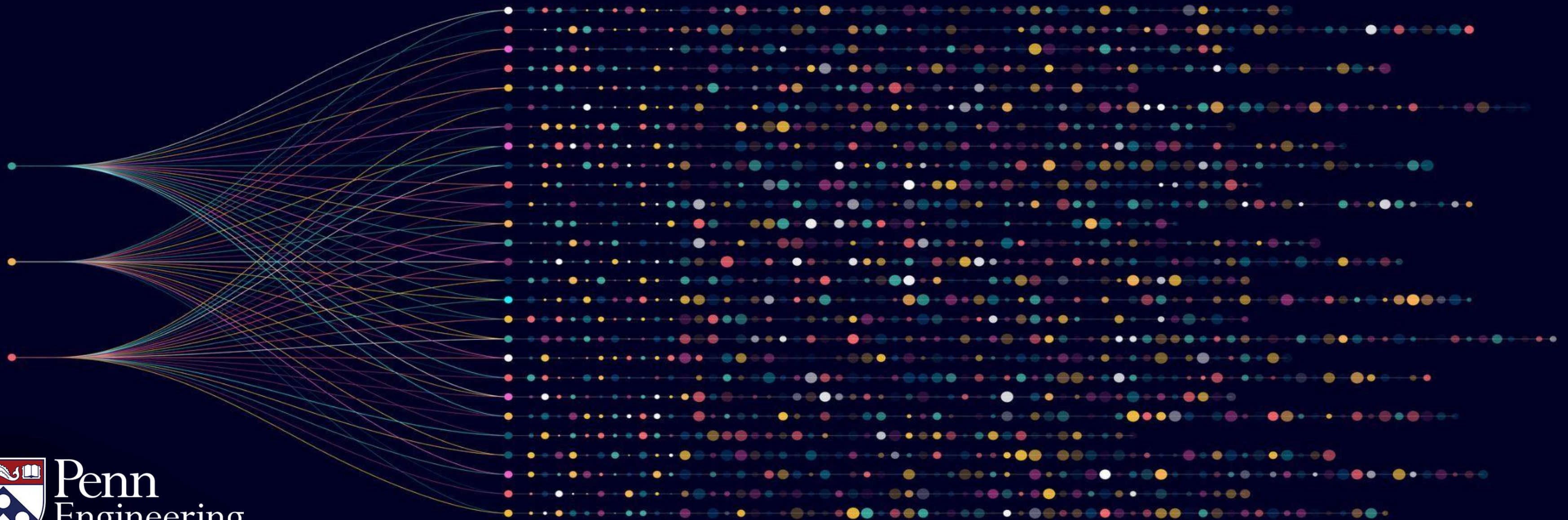


Algorithms for Big Data

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Variance and Chebyshev's inequality



An anecdote about averages

Famous computer scientist and Columbia CS Professor Christos Papadimitriou has ~ 300 co-authors. Average worth of his co-authors is nearly 400 million dollars.



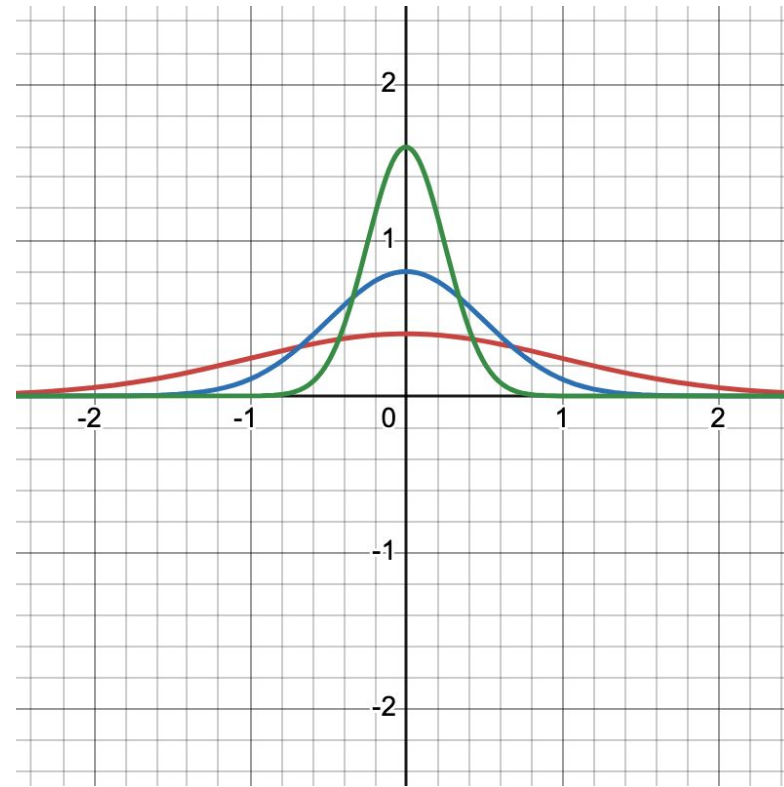
An anecdote about averages

Christos wrote a paper in the mid 70s with a then undergraduate student – Bill Gates (today worth ~ \$120 billion)



Averages can be misleading!

Moral of the story: averages can be misleading and do not tell the entire story for a distribution



Red curve has maximum spread and the **green** curve has minimum spread

Variance of a random variable

Given a real valued random variable $\mathbf{X}$, its variance is defined as

$$\text{Var}(\mathbf{X}) := \text{E}[(\mathbf{X} - \mu)^2] \text{ where } \mu := \text{E}[\mathbf{X}].$$

The variance measures the “average squared distance” of a random variable from its expectation

Note that $\text{Var}(\mathbf{X}) \geq 0$. In fact, the only way $\text{Var}(\mathbf{X}) = 0$ if $\mathbf{X}$ only takes one value.

Alternate way of writing variance

$Var(\mathbf{X}) := E[(\mathbf{X} - \mu)^2]$ where $\mu := E[\mathbf{X}]$.

$$\begin{aligned} E[(\mathbf{X} - \mu)^2] &= E[\mathbf{X}^2 + \mu^2 - 2\mu\mathbf{X}] \\ &= E[\mathbf{X}^2] + \mu^2 - 2E[\mu\mathbf{X}] \text{ (by linearity of expectation)} \quad - (1) \end{aligned}$$

Now, $E[\mu\mathbf{X}] = \mu E[\mathbf{X}] = \mu^2$. Plugging this back into (1), we get
 $E[(\mathbf{X} - \mu)^2] = E[\mathbf{X}^2] + \mu^2 - 2\mu^2 = E[\mathbf{X}^2] - \mu^2$.

Thus, $Var(\mathbf{X}) = E[\mathbf{X}^2] - \mu^2$ -- i.e.,
“gap between the expectation of the square and the square of the expectation”

Corollary: Since $Var(\mathbf{X}) \geq 0$, it follows that $E[\mathbf{X}^2] \geq (E[\mathbf{X}])^2$

Small variance $\approx$ Concentration of data

As $Var(\mathbf{X}) := E[(\mathbf{X} - \mu)^2]$, at a high level, $Var(\mathbf{X})$ measures how far $\mathbf{X}$ is from its mean

So, at an intuitive level, the smaller $Var(\mathbf{X})$ is, the more $\mathbf{X}$ is concentrated around its mean

Can we make this formal?

Chebyshev's inequality

Chebyshev's inequality: For any random variable $\mathbf{X}$,

$$\Pr[|\mathbf{X} - \mathbf{E}[\mathbf{X}]| \geq t \sigma] \leq \frac{1}{t^2},$$

where $\sigma^2 = \text{Var}(\mathbf{X})$. The quantity $\sigma = \sqrt{\text{Var}(\mathbf{X})}$ is referred to as the standard deviation of $\mathbf{X}$.

Intuitively: The standard deviation of $\mathbf{X}$, i.e., $\sqrt{\text{Var}(\mathbf{X})}$, determines the “scale” at which $\mathbf{X}$ typically deviates from its mean

As an example: If we plug in $t = 5$, the probability that $\mathbf{X}$ deviates from $\mathbf{E}[\mathbf{X}]$ by more than 5σ is at most $\frac{1}{t^2} = \frac{1}{5^2} = 4\%$

Proof of Chebyshev's inequality

Consider the random variable $\mathbf{Z} = (\mathbf{X} - \mathbf{E}[\mathbf{X}])^2$.

By definition, $\mathbf{E}[\mathbf{Z}] = \text{Var}(\mathbf{X}) := \sigma^2$.

Since $\mathbf{Z}$ is a non-negative random variable, by Markov's inequality, for any $t \geq 0$,

$$\Pr[\mathbf{Z} \geq t^2 \sigma^2] \leq \frac{\mathbf{E}[\mathbf{Z}]}{t^2 \sigma^2} = \frac{\sigma^2}{t^2 \sigma^2} = \frac{1}{t^2} \quad - (1)$$

Observe that $\mathbf{Z} \geq t^2 \sigma^2$ if and only if $|\mathbf{X} - \mathbf{E}[\mathbf{X}]| \geq t \sigma$.

Thus, $\Pr[|\mathbf{X} - \mathbf{E}[\mathbf{X}]| \geq t \sigma] = \Pr[\mathbf{Z} \geq t^2 \sigma^2] \leq \frac{1}{t^2}$ (using (1)).

Proof of Chebyshev's inequality: in retrospect

Chebyshev's inequality is just Markov's inequality on the random variable $\mathbf{Z} = (\mathbf{X} - \mathbf{E}[\mathbf{X}])^2$

Many probabilistic bounds are obtained by Markov's inequality on the “right random variable”. More examples later.

Summary

Variance is the gap between the "expectation of the square" and "the square of the expectation"

Variance (and standard deviation) are measures of the spread of a random variable

Chebyshev's inequality implies (for example) that a random variable deviates more than 5σ from its mean is at most 4%